

Trustworthy and Explainable Agents for Phishing Detection with Grounded Cross-Model Evidence Agreement

Abstract

Large language models (LLMs) now flag phishing websites with high accuracy, but a deployed detector consumes a *label* and often discards the *reason*. For actions such as browser warnings, blocklist entries, and take-down requests, operators need a reason they can check and a criterion for when to trust the verdict automatically. Label confidence and majority voting miss a common failure: several models can agree on the right verdict while relying on different, or even fabricated, evidence. We propose PHISHPROOF, a multi-agent selective detector that reads reliability from grounded evidence agreement rather than from labels. A small panel of diverse LLM evidence agents investigates a page, plans which URL, DOM, certificate, redirect, screenshot, and logo signals to inspect, grounds each claim with verification tools, and writes the results into a shared typed evidence graph. PHISHPROOF acts only when the agents agree on a subgraph that the page itself confirms; otherwise it abstains, returning the agreed verified subgraph as the explanation when it does act. A delayed blocklisting oracle calibrates the score to a probability of correctness. On a six-month phishing benchmark, PHISHPROOF lowers AURC by 35% relative to the strongest label-agreement baseline, raises coverage at 99% selective accuracy from 0.48 to 0.71, and abstains on 78–83% of the right-label, wrong-reason and ungrounded regions that baselines act on confidently.

Keywords

phishing detection, LLM agents, tool-augmented agents, context graphs, selective prediction, cross-model evidence agreement, explanation faithfulness, abstention, trustworthy ML

1 Introduction

A phishing website is a consequential object: when a detector marks a page as phishing, the verdict drives a blocklist entry, a browser warning, or a take-down request that removes a live service from the web. Large language models have raised the accuracy of such detectors sharply, reading a page’s URL, DOM, rendered screenshot, and certificate to decide whether it impersonates a brand it does not own [13, 15]. Yet accuracy is only half of what a deployment needs. The analyst who acts on a verdict, the operator who must justify a take-down, and the user who sees a warning all need a *reason* they can check – and they need to know *when* the detector is trustworthy enough to act on automatically. For a decision that can be contested, the bottleneck is no longer whether the model is usually right, but whether a particular verdict can be *trusted and explained*.

Existing detectors approach this only indirectly. One line runs a single LLM over the page and reports a class probability or a verbalized confidence [15]; reliability is then a property of that one model’s calibration [7]. A second line aggregates several models or several samples – self-consistency over one model [24], majority voting over an ensemble [22], or weighted label agreement in the

style of crowd-sourcing and label-free evaluation [4, 5] – and reads reliability off how often the votes concur. A third, recent line builds multi-agent detectors whose agents specialize on different signals and whose rationales are merged into a single explanation [12, 25]. Together these families cover much of the space of accuracy and aggregation.

They share a blind spot, visible in a routine case (Fig. 1). A page impersonating a bank is shown to a panel of three diverse models; all three return “phishing,” so every aggregation method above reports high confidence. But the models reached that verdict for different reasons: one cited the morphed logo, one cited a credential form posting off-domain, and one cited a certificate mismatch that does not actually hold on the page. First, the detectors emit a label and discard these reasons, so nothing auditable reaches the operator (L1). Second, reliability is read at the *label*, where the three agree, and is therefore blind to the fact that they *disagree on the evidence* – the verdict is right for the wrong reason (L2). Third, where the models do agree on a reason, that agreement can be a shared hallucination of evidence that is not on the page (L3). Fourth, even a detector that wanted to gate on agreement has no ground-truth at inference time to calibrate how much agreement is enough, and the threshold it picks drifts as campaigns evolve (L4). A page that is right for the wrong reason is not a curiosity: it is exactly the page a small change to the cited cue would flip, and exactly the page an operator should not act on blindly.

Two properties of phishing, not shared by generic question answering, make a direct remedy possible now. The evidence a detector relies on is *objectively checkable*: a form-action domain, a certificate organization, a redirect target, and a rendered logo can each be re-derived from the page itself. And phishing carries a delayed but high-precision *oracle* – a malicious URL is later blocklisted or taken down [18, 20] – that supplies real labels after the fact. Crucially, the evidence that decides a phishing verdict is *heterogeneous*: it spans several model types (text LLMs and a vision-language model), many signals (the URL, DOM, certificate, redirect chain, logo, and credential intent), two modalities, and the verification mechanisms that confirm each one. Deciding which of these to gather for a given page and how to combine them is a *planning-and-aggregation* problem that suits an agent rather than a fixed pipeline – and modern LLMs can act as such agents, emitting typed outputs and calling tools to gather and verify their own evidence. The opportunity is to stop treating reliability as a property of labels and start treating it as a property of *evidence that several independent agents plan, gather, verify, and agree on, and that the page confirms*.

To this end, we present PHISHPROOF, a multi-agent phishing detector that trusts a verdict in proportion to grounded cross-model agreement on the evidence. A small, deliberately diverse panel of LLM *evidence agents* investigates the page; rather than emit a bare label, each agent *plans* which signals and modalities to examine, gathers the evidence and grounds it by calling verification *tools* – a DOM parser, a certificate reader, a redirect tracer, and a logo

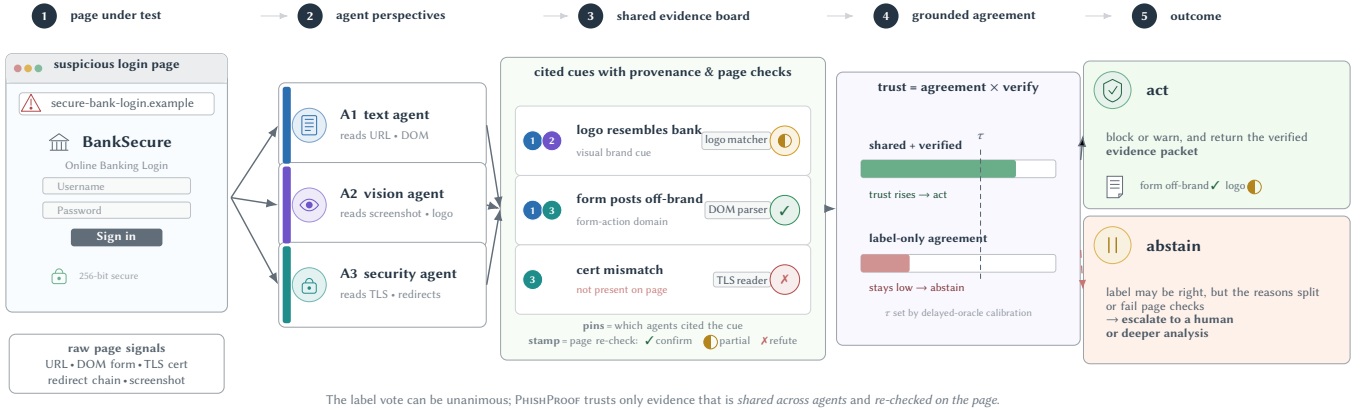


Figure 1: PHISHPROOF as a qualitative evidence story. Three agents inspect the same page from complementary views, then place their cited cues on a shared evidence board. Colored pins show provenance, while tool badges mark cues that are confirmed or rejected by the page itself. PHISHPROOF acts when the shared reasons survive these checks and returns them as an evidence packet; when agents agree on the label but their reasons split or fail verification, it abstains.

matcher – then records what it found in a shared, per-page *evidence graph* (a context graph) whose typed nodes carry the asserting agent and the confirming tool. PHISHPROOF *aggregates* the agents’ grounded findings into the graph, measures agreement at the level of the *graph* rather than the label, weights it by whether each agreed node verifies against the page, and combines the two into a single Grounded Evidence-Agreement score. It acts only when that score is high – returning the agreed, verified subgraph as a navigable, decision-aware explanation – and abstains otherwise, escalating the page to deeper analysis or a human. A delayed-oracle calibrator maps the score to a probability of correctness so the operating point is set against real outcomes rather than guessed. The novelty over prior multi-agent detectors is the unit of agreement: not the label, and not one model’s introspective faithfulness, but a shared evidence graph that is simultaneously cross-agent-agreed and tool-verified.

Realizing this raises four difficulties that the design must rule out. Agreement over structured evidence must be defined so that it is not label agreement in disguise – two models can name the same brand while resting on different cues. Shared hallucination must be caught cheaply, without trusting the models to police themselves. The trust score must be calibrated without any inference-time label and must degrade gracefully as campaigns drift, using only the oracle’s late arrivals. And the agent panel must stay small, since every added agent is a round of model calls and tool calls, so the gain has to come from *diversity* rather than count.

We make the following contributions.

- **Failure mode.** We identify the *right-label, wrong-reason* failure of multi-model phishing detectors and cast trust-worthy detection as selective prediction over an evidence-agreement signal (§3).
- **Method.** We introduce PHISHPROOF, a panel of tool-using evidence agents that build a shared per-page evidence graph, and its Grounded Evidence-Agreement score, which measures cross-agent agreement over that graph and turns it,

via delayed-oracle calibration, into a selective act/abstain decision with the agreed subgraph as the explanation (§4).

- **Evidence.** On a six-month phishing benchmark, PHISHPROOF improves the risk–coverage trade-off over single-model confidence, self-consistency, and label voting: AURC falls by 35% relative to the strongest baseline and coverage at 99% selective accuracy rises from 0.48 to 0.71 (§6).
- **Analysis.** We isolate the contribution of evidence-level agreement, tool-grounding, agent diversity, and calibration. The right-label, wrong-reason and ungrounded regions flip under evidence-targeted perturbations 41.6% and 47.9% of the time, and PHISHPROOF abstains on 78–83% of them (§6).

The rest of the paper is organized as follows. §2 positions PHISHPROOF against label-level reliability and multi-agent detection. §3 formalizes grounded evidence-agreement and the selective objective. §4 details the evidence agents, the graph they build, the score, and the calibrated decision rule. §5 discusses deployment considerations, and §6 reports the evaluation.

2 Related Work

LLM and reference-based phishing detection. The dominant modern detectors decide whether a page impersonates a brand it does not own. Reference-based systems match a rendered page against a brand library of logos and layouts [1, 14, 16], and recent work refreshes that reference set online [17] or replaces it with an LLM that recognizes the brand and checks the domain directly [13, 15]. A parallel line treats the page as URL or image features for a learned classifier [26], surveyed broadly [21]. These systems optimize and report *aggregate* accuracy. Departing from this line, PHISHPROOF adds no new base detector and makes no accuracy claim; it is a layer that decides, *per verdict*, whether the panel’s decision can be trusted and acted on (§4.4).

Multi-agent and ensemble phishing detection. Closest to us are detectors that run several models. Majority voting over LLMs has been studied directly for phishing [22]; multi-agent systems assign

agents to different signals and either debate to a verdict or merge their rationales into one explanation [12, 25], sometimes backed by an episodic memory of past cases [3]. Unlike these systems, which aggregate or consolidate at the level of the *label* or a fused rationale, PHISHPROOF measures agreement at the level of individually cited evidence units ((2)) and treats *disagreement on evidence* as the signal to abstain rather than a conflict to resolve. Its agents are tool-using investigators that each *plan* their own investigation across signals and modalities, and whose tool-grounded findings are *aggregated* into a per-page evidence (context) graph that serves as both the trust signal and the explanation – in the spirit of decision-aware agents that reason over an explicit graph – rather than a fused free-text rationale.

Label-level reliability and label-free evaluation. A verdict’s reliability is usually read from the label distribution: post-hoc confidence calibration [7], agreement across samples of one model [24], or agreement across many models, as in crowd-sourcing aggregation [4], agreement-on-the-line under shift [2], and trust calibration for phishing detectors [5]. Where these estimate reliability from *how often labels concur*, PHISHPROOF estimates it from *whether the reasons concur and hold*, which separates pages on which the panel agrees by coincidence from those on which it agrees because the page truly carries the evidence ((4)).

Explanation faithfulness. Causal audits show that an LLM’s stated rationale is often not the driver of its decision [9], and multi-agent debate has been used to *improve* the faithfulness of generated explanations [10]. Such audits operate on a single model and require counterfactual re-execution. In contrast, PHISHPROOF uses cross-model agreement on grounded evidence as an inexpensive faithfulness proxy: a reason that several independent models cite and that the page confirms is, by construction, less likely to be a post-hoc rationalization, at no extra re-execution cost.

Selective prediction. We build on selective classification, which trades coverage for accuracy via an abstention rule [6], and on distribution-free uncertainty quantification [23]. Building on this foundation, PHISHPROOF supplies the missing ingredient for the phishing setting: an abstention score derived from grounded cross-model agreement, and a calibration target – the delayed blocklisting oracle – that lets the risk-coverage operating point be set against real outcomes rather than a held-out guess.

Position and scope. PHISHPROOF contributes a single new object, the Grounded Evidence-Agreement score, and the selective detector built on it. Tab. 1 positions it against one representative per family along five capabilities – evidence-level agreement, page-grounding, oracle calibration, selective abstention, and a verifiable explanation as output – and PHISHPROOF is the only entry that supports all five. We focus on per-verdict trust and explanation for a multi-model phishing detector; the framework is agnostic to the specific base models and evidence schema. We do *not* claim (i) a new base detector or higher full-coverage accuracy – PHISHPROOF layers over any panel; (ii) robustness to an adversary who controls the panel agents or corrupts the tools’ inputs, a harness-level threat we treat as out of scope; or (iii) coverage guarantees on fully cloaked, novel pages for which the oracle has not yet arrived, where the detector necessarily abstains rather than errs.

Table 1: Capability comparison for per-verdict trust in multi-model phishing detection. ●: yes; ◐: partial; ○: no. Columns are defined in the body.

Approach	Evid.- level	Page- grounded	Oracle- calib.	Selective abstain	Explan. output
Single conf. [7]	○	○	◐	●	○
Self-consist. [24]	○	○	○	◐	○
Majority vote [22]	○	○	○	◐	○
Label agree. [4]	○	○	◐	◐	○
MultiPhishGuard [25]	◐	○	○	◐	◐
Faithfulness audit [9]	◐	○	○	○	◐
PHISHPROOF (ours)	●	●	●	●	●

3 Problem Formulation

Setting. A detector receives a webpage x , represented by its URL, DOM, rendered screenshot, TLS certificate, and redirect chain, and must decide whether x is phishing. We assume access to a panel $\mathcal{M} = \{m_1, \dots, m_M\}$ of M diverse LLM evidence agents (different model families, including at least one vision-language model that reads the screenshot), each able to call a set of verification tools, and to a fixed, typed evidence schema $\mathcal{U}$. A unit $u = (t, v) \in \mathcal{U}$ pairs a type t (e.g. form-action-domain, cert-org, redirect-target, logo-brand, credential-intent) with a value v ; each type has a tool g that re-derives it from x . The agents’ findings are assembled into a per-page evidence graph $H(x)$ (a context graph) whose nodes are grounded units and whose edges record the brand-versus-infrastructure relations among them. Notation is summarized in Tab. 2.

Agent outputs and the evidence graph. Run on x , each agent investigates the page and returns a verdict together with the evidence nodes it grounds through tool calls:

$$m_i(x) = (y_i(x), E_i(x)), \quad y_i \in \{\text{phish}, \text{benign}\}, \quad E_i \subseteq \mathcal{U}. \quad (1)$$

The agents’ nodes form the page’s evidence graph $H(x)$, with node set $\bigcup_i E_i(x)$ and edges linking each brand claim to the infrastructure nodes that support or contradict it. The verdict is the majority vote $\hat{y}(x) = \text{maj}_i y_i(x)$. A *consensus* node is one a strict majority of the agents assert with a consistent value, $C(x) = \{u : |\{i : u \in E_i\}| > M/2\}$; these consensus nodes are the subgraph PHISHPROOF trusts and returns as the explanation.

Prior formulation and its limits. Existing detectors estimate a per-verdict reliability $r(x)$ from the *label* channel alone – a single model’s confidence, or the concordance of $\{y_i\}$ [4, 7, 22]. This label-centric formulation has four limitations, which we restate as the requirements PHISHPROOF must meet (Tab. 3). It returns no checkable artifact, only $\hat{y}$, so an action cannot be justified (**L1**; **R1**: emit structured evidence). It is a function of $\{y_i\}$ only, so it cannot distinguish a panel that agrees *because* the page carries the evidence from one that agrees on the label while resting on different cues (**L2**; **R2**: score agreement on evidence). It treats a cited reason as true, so a reason all models hallucinate is rewarded (**L3**; **R3**: ground evidence to the page). And it offers no target against which to set its operating point as campaigns drift (**L4**; **R4**: calibrate against the delayed oracle).

Table 2: Notation.

Symbol	Meaning
x	a webpage under test (URL, DOM, screenshot, cert, redirects)
$\mathcal{M} = \{m_1, \dots, m_M\}$	panel of M diverse LLM evidence agents ($M=3$)
$\mathcal{U}$	fixed typed evidence-unit schema (§4.1)
$H(x)$	per-page evidence graph (context graph) the agents build
$E_m \subseteq \mathcal{U}$	grounded nodes agent m contributes on x
C	consensus subgraph (nodes a majority of agents assert)
$g(u, x) \in [0, 1]$	tool that re-derives unit u 's type from x
$A(x) \in [0, 1]$	evidence-level agreement over the graph
$G(x) \in [0, 1]$	groundedness of the consensus nodes
$\text{GEA}(x) = A \cdot G$	Grounded Evidence-Agreement score
τ	abstention threshold (acts iff $\text{GEA} \geq \tau$)
$\hat{\rho} = \kappa(\text{GEA})$	calibrated map $\text{GEA} \rightarrow \mathbb{P}(\text{correct})$

Table 3: The traceable thread: each limitation (§1) becomes a requirement and a named component.

Limitation	Requirement	Component
L1 label-only, unverifiable output	R1 emit structured, checkable evidence	C_1 Agentic evidence elicitation (§4.1)
L2 reliability read at the label, blind to wrong reasons	R2 measure agreement at the evidence level	C_2 Evidence agreement A (§4.2)
L3 shared hallucination of false evidence	R3 ground cited evidence to the page	C_3 Tool-grounding G (§4.3)
L4 no ground-truth calibration; drift	R4 calibrate & select against the delayed oracle	C_4 Calibrated selection (§4.4)

Grounded evidence-agreement. We replace label reliability with a score on the *evidence* channel. Define the evidence-level agreement and the groundedness of the consensus,

$$A(x) = \frac{|C(x)|}{|\bigcup_i E_i(x)|}, \quad (2)$$

$$G(x) = \frac{1}{|C(x)|} \sum_{u \in C(x)} g(u, x), \quad (3)$$

where $g(u, x) \in [0, 1]$ is the tool that re-derives unit u 's type from x (deterministic for form-action, cert-org, redirect; perceptual for logo-brand); $G(x) = 0$ when $C(x)$ is empty. The *Grounded Evidence-Agreement* score is their product,

$$\text{GEA}(x) = A(x) \cdot G(x) \in [0, 1]. \quad (4)$$

Agreement on units the page does not confirm ($G \rightarrow 0$) and label agreement without evidence consensus ($A \rightarrow 0$) both drive GEA down, directly discharging **R2** and **R3**.

Objective. PHISHPROOF is a selective detector: it acts with $\hat{y}(x)$ when $\text{GEA}(x) \geq \tau$ and abstains ($\perp$) otherwise. For a threshold τ , coverage and selective risk are

$$\phi(\tau) = \mathbb{P}[\text{GEA} \geq \tau], \quad r(\tau) = \mathbb{P}[\hat{y} \neq y \mid \text{GEA} \geq \tau], \quad (5)$$

with y the true label. We seek a score whose risk-coverage curve $r(\phi)$ lies as low as possible (small area under it, AURC), and a calibrator $\kappa : \text{GEA} \mapsto \hat{\mathbb{P}}(\hat{y} = y)$ fit on the delayed oracle so that the operating point τ realizes a target risk (**R4**). *Outputs* per page: a verdict or abstention, the calibrated trust $\hat{\rho} = \kappa(\text{GEA})$, and, when acting, the consensus subgraph $C(x)$ as the explanation.

Framework overview. Fig. 2 maps the requirements to four components evaluated in §6: agentic evidence elicitation – agents call tools to build the graph (C_1 , **R1**), evidence agreement over the graph (C_2 , **R2**), tool-grounding (C_3 , **R3**), and calibrated selection (C_4 , **R4**). §4 details each; Tab. 3 is the thread we follow through to the ablation.

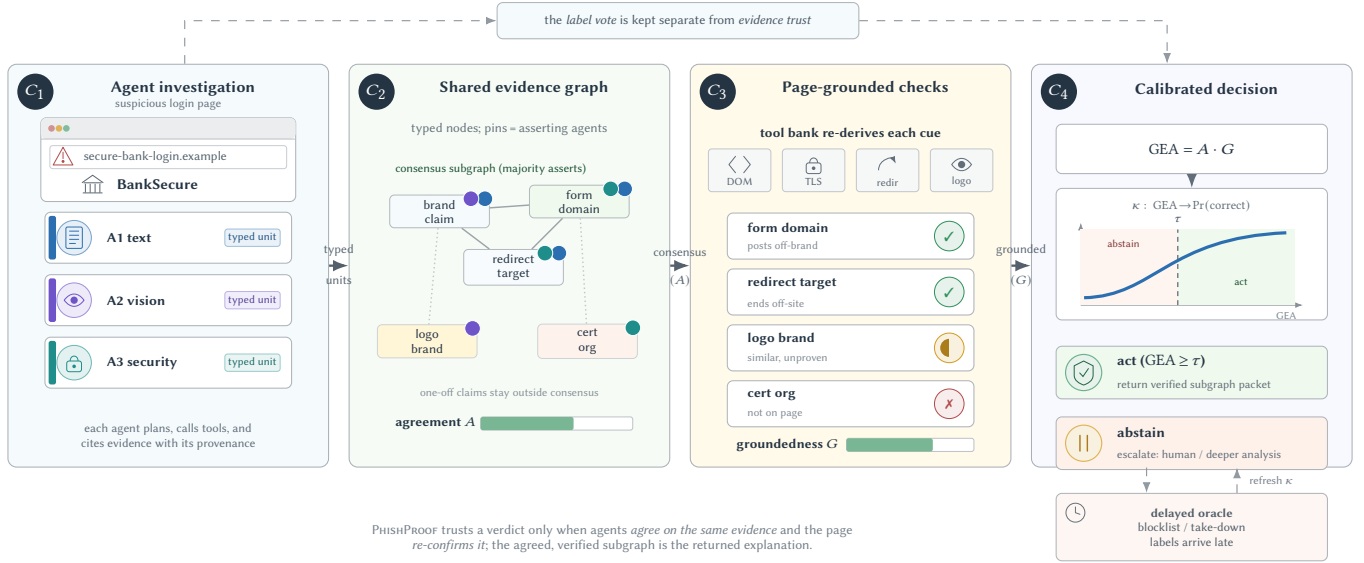


Figure 2: PHISHPROOF qualitative pipeline. Agents inspect the same page from complementary views, leave cited evidence notes, and place those notes on a shared evidence board. The system trusts a verdict only when the agents converge on the same page cues and the tool bank can re-check those cues against the live page. The calibrated selector returns the label with a verified evidence packet when the evidence is shared and grounded; otherwise it abstains, including cases where the label vote is plausible but the reasons are thin or unverified.

4 The PHISHPROOF Detector

PHISHPROOF turns the score of (4) into a calibrated, selective verdict through the four components of Fig. 2. We describe each, then give the end-to-end procedure and its cost.

4.1 Evidence agents and the context graph

Each member of the panel is an *evidence agent*: prompted with the page, it *plans* which signals and modalities to examine, calls the verification tools to obtain them, and returns a verdict together with the grounded evidence nodes it found, drawn from the fixed schema $\mathcal{U}$ and emitted by constrained decoding as well-typed (t, v) pairs rather than free text (R1). The schema is small and auditable – six types covering the brand claim, the form-action domain, the certificate organization, the redirect target, credential intent, and the logo brand – and is shared by every agent, so their nodes are directly comparable. The agents’ nodes are assembled into the page’s *evidence graph* $H(x)$ (a context graph): nodes are typed, tool-grounded evidence units tagged with the agent that asserted them, and edges link each brand claim to the infrastructure nodes that support or contradict it, so the verdict is a path through the graph – a brand node joined to off-brand infrastructure – rather than an opaque score. Agent *diversity* is a design choice, not an afterthought: the three agents come from different model families and at least one is a vision-language model that reads the screenshot, so that agreement on the graph reflects independent evidence rather than a shared prior (its value is isolated in §6).

4.2 Evidence agreement

To *aggregate* and compare nodes across agents we first *normalize values*: domains are reduced to their registrable eTLD+1, brands to a canonical lexicon entry, so that “paypal-secure.com” and “paypal-secure.com/login” count as the same form-action node. The consensus subgraph $C(x)$ collects the value-normalized nodes a strict majority of the agents assert, and the evidence-level agreement $A(x)$ of (2) is the share of all nodes in the graph that fall in this consensus (R2). Agreement is thus high only when the agents converge on the *same* evidence, and is unaffected by their concurring on the label alone.

4.3 Tool-grounding

A consensus can still be wrong if the agents share a hallucination, so each consensus node is checked against the page by the tool for its type $g(u, x)$ (R3). The structural tools verify deterministically – the DOM is parsed for the form-action domain, the certificate for its organization, and the redirect chain for its final target – and return 0 or 1. The perceptual tool (logo-brand) matches the rendered logo to the claimed brand and returns a similarity in $[0, 1]$. The groundedness $G(x)$ of (3) averages these over the consensus, and the score $\text{GEA} = A \cdot G$ of (4) collapses whenever the agreed evidence does not survive verification.

4.4 Calibrated selection

Finally, PHISHPROOF turns the score into a decision (R4). Because phishing labels arrive late – a malicious URL is eventually block-listed or taken down [18, 20] – we fit an isotonic calibrator $\kappa : \text{GEA} \mapsto \hat{\mathbb{P}}(\hat{y} = y)$ on pages whose oracle label has already arrived,

Algorithm 1 PHISHPROOF inference on a page x

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1: for  $m_i \in \mathcal{M}$ :  $(y_i, E_i) \leftarrow m_i(x)$  ▷  $C_1$  agents call tools, (1)
2:  $\hat{y} \leftarrow \text{maj}_i y_i$ ;  $H \leftarrow \bigcup_i E_i$  ▷ evidence graph
3:  $C \leftarrow \{u : u \text{ asserted by } > M/2 \text{ agents, value-normalized}\}$ 
4:  $A \leftarrow |C| / |\bigcup_i E_i|$  ▷  $C_2$ , (2)
5: for  $u \in C$ : compute  $g(u, x)$ ;  $G \leftarrow \text{mean}_{u \in C} g(u, x)$  ▷  $C_3$ , (3)
6:  $\text{GEA} \leftarrow A \cdot G$ ;  $\hat{p} \leftarrow \kappa(\text{GEA})$  ▷  $C_4$ , (4)
7: if  $\text{GEA} \geq \tau$  return  $(\hat{y}, \hat{p}, C)$ ; else return  $\perp$ 

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and refresh it on a sliding window so the mapping tracks drift. The detector acts with $\hat{y}(x)$ when $\text{GEA}(x) \geq \tau$, where τ is the smallest threshold meeting a target selective risk under κ , and abstains otherwise, escalating the page to deeper analysis or a human. When it acts, it returns the consensus subgraph $C(x)$ as a verifiable explanation.

4.5 Procedure, guarantee, and cost

Walk-through. 1 runs the agent panel once, each agent calling its tools to build the evidence graph (line 1), takes the majority label (line 2), and then does the work that distinguishes PHISHPROOF from label-level reliability: it forms the consensus subgraph (line 3), scores agreement on that evidence rather than on the label (line 4), checks each agreed node against the page with its tool so a shared hallucination cannot inflate the score (line 5), and combines the two factors into GEA (line 6). The calibrated trust and the selective decision (line 7) set the operating point against real outcomes. The novelty is concentrated in lines 3–6: the quantity that gates the verdict is grounded agreement on the evidence *graph*, and the returned subgraph is simultaneously the trust signal and the explanation.

Guarantee. Tool-grounding is what rules out shared hallucination, and we can state this precisely.

ASSUMPTION 1 (TOOL SOUNDNESS). *For each unit type, the tool $g(u, x) = 1$ if u holds on x and 0 otherwise (perceptual types: $g(u, x)$ is the true match score).*

PROPOSITION 1. *Under 1, any consensus node that does not hold on x contributes 0 to $G(x)$; hence $\text{GEA}(x)$ is at most the grounded fraction of the consensus, and $\text{GEA}(x) = 0$ whenever the entire consensus is hallucinated.*

PROOF SKETCH. G is the mean of g over C ((3)); by 1 a non-holding node adds 0 to that mean, so $G \leq |\{u \in C : u \text{ holds}\}|/|C|$, and $\text{GEA} = A \cdot G \leq G$ inherits the bound. If no consensus node holds, $G = 0$ and thus $\text{GEA} = 0$. $\square$

The proposition is modest by design: it bounds, but does not eliminate, gauge error, since a tool is sound for what it checks but a page can carry true-yet-misleading evidence; we report tool precision and recall against gold in §6.

Complexity. Per page, PHISHPROOF costs M agents – each a model call plus the tool calls it issues – and the verification of the $|C|$ consensus nodes; the structural tools are $O(1)$ DOM/certificate/redirect lookups and the single perceptual check is one embedding comparison, so the cost is dominated by the $M=3$ agents and is independent of the schema size beyond the nodes actually asserted. Relative to a single-model detector this is a fixed $M\times$ overhead with no training;

under selective operation only the abstaining fraction $1 - \phi(\tau)$ is escalated to deeper analysis, so the amortized cost is tuned by the same threshold that sets the risk–coverage operating point (§6).

5 Further Considerations

Each consideration below is handled by reusing a component of §4, not by adding a new one.

Unseen brands and the long tail. A page may impersonate a brand outside the canonical lexicon, where the value-normalization of §4.2 cannot resolve a logo-brand or brand-claim unit. In that case the score is still defined on the structural units the page does carry – form-action, cert-org, and redirect require no brand lexicon and remain verifiable through (2)–(3). The trade-off is lower evidence coverage: unresolvable brand units drop out of the consensus, so the detector may abstain more often. We treat the resulting long-tail coverage as a future-work measurement, not as a claim of this evaluation.

Bounded panels and budget. Every agent is a round of model and tool calls, so the M -agent cost is the binding budget constraint. The selective rule of §4.4 already bounds it: only the abstaining fraction is escalated, so the average cost is set by the same threshold τ that fixes the risk–coverage operating point. When budget is tight a two-stage variant runs a single agent first and convenes the full panel only on its low-confidence pages; this reuses the abstention machinery unchanged and adds agent passes on an $O(1-\phi)$ fraction of traffic rather than on all of it.

New evidence types and channels. The schema $\mathcal{U}$ is open. A new signal – a unit drawn from the surrounding message-channel text when the URL arrives over chat, or from a community scam report – is admitted by registering one additional type and its tool g ; the agreement and groundedness of (2)–(3) are defined over whatever types are present and need no change, and the calibrator re-fits on the same oracle. This is an interface claim rather than an empirical coverage claim: each new channel still needs its own validation set, verifier, and cost measurement before it should be counted as evaluated.

6 Experiments

We evaluate PHISHPROOF as a selective detector: it may act only on pages whose evidence graph is trustworthy enough, and should abstain otherwise. The five questions below each test one method claim or deployment constraint from §3:

- **RQ1 – Selective detection (§6.2):** does grounded evidence-agreement give a better risk–coverage trade-off than label-level reliability?
- **RQ2 – Right-label, wrong-reason (§6.3):** does PHISHPROOF flag the pages a panel labels correctly but for the wrong reason, which the baselines act on?
- **RQ3 – Component attribution (§6.4):** how much do evidence-level agreement, groundedness, diversity, and calibration each contribute?
- **RQ4 – Calibration and drift (§6.5):** is the trust score calibrated, and does it hold across a temporal split?
- **RQ5 – Cost and scope (§6.6):** what does the panel cost under selective operation, and which practical extensions remain outside the measured evaluation?

6.1 Setup

Benchmark and oracle. We evaluate on PHISHSEL, a benchmark of roughly 4,000 webpages assembled over a six-month window: live phishing URLs drawn from public feeds [18, 20] and benign pages sampled from the Tranco top-sites list [11], each captured with its URL, DOM, rendered screenshot, TLS certificate, and redirect chain. Ground truth for evaluation comes from the delayed oracle – a phishing URL’s eventual blocklisting or take-down – recorded after a fixed lag, which also supplies the labels the calibrator of §4.4 consumes. The six-month span lets us hold out a later slice for the temporal study (RQ4).

Agents, tools, and baselines. The default panel is $M=3$ deliberately diverse *evidence agents*: a frontier text LLM, a second text LLM from a different family, and a vision-language model that reads the screenshot, each able to call the verification tools of §4.3. All methods receive the same captured page snapshot and no oracle label at inference time. PHISHPROOF emits constrained evidence units in the schema of §4.1, normalizes values as in §4.2, and verifies only the consensus units before calibration. We compare against six reliability baselines that share this panel or a member of it:

- *B1 Single-model confidence* [7]: the strongest single model with temperature-scaled softmax confidence.
- *B2 Verbalized confidence*: the same model asked to state its confidence.
- *B3 Self-consistency* [24]: one model, $k=5$ samples, agreement as the score.
- *B4 Majority vote* [22]: panel label agreement.
- *B5 Label-agreement* [4]: reliability-weighted label agreement over the panel.
- *B6 MultiPhishGuard* [25]: a multi-agent detector whose consolidated-rationale confidence is used as the score.

Tasks and metrics. All methods produce a per-page trust score and we evaluate them as selective detectors via (5). We report the area under the risk–coverage curve (*AURC*, selective risk integrated over coverage); selective accuracy and false-positive rate at 80% coverage (*SelAcc*₈₀, *FPR*₈₀); the coverage attainable at 99% selective accuracy (*Cov*₉₉); and the expected calibration error of the trust score (*ECE*). For RQ2 we also report the *flip rate*: the fraction of pages whose panel label changes under a small evidence-targeted perturbation (a logo morph [8] or a form-action cloak).

Protocol. Thresholds for *SelAcc*₈₀ and *Cov*₉₉ are chosen on a calibration split and then applied unchanged to the held-out evaluation pages. The isotonic calibrator uses only pages whose delayed oracle label has arrived; for the temporal protocol we fit it on months 1–3 and evaluate on months 4–6. For RQ2, perturbations target the cited evidence cue while preserving the page label, so a flip measures reason fragility rather than a changed ground truth.

Statistical reporting. Numeric cells report mean_{±95% CI} over five seeds, with confidence intervals from a paired bootstrap over pages. We treat non-overlapping intervals as evidence of a reliable difference and state relative gains only for comparisons whose intervals do not overlap. The proposed method is **bold**; the best baseline is underlined.

Run environment. The text models are queried through their providers’ APIs at temperature 0 with the same five seed values

Table 4: RQ1. Selective detection on PHISHSEL. Lower AURC/FPR/ECE and higher SelAcc/Cov are better. Cells are mean_{±95% CI} over five seeds; best, best baseline. The Δ row is relative to B5.

Method	AURC↓	SelAcc ₈₀ ↑	FPR ₈₀ ↓	Cov ₉₉ ↑	ECE↓
B1 Single-model conf.	5.9 _{±0.4}	95.8 _{±0.4}	4.1 _{±0.5}	0.38 _{±0.05}	0.082 _{±0.009}
B2 Verbalized conf.	6.8 _{±0.5}	95.0 _{±0.5}	4.9 _{±0.6}	0.29 _{±0.05}	0.110 _{±0.011}
B3 Self-consistency ($k=5$)	5.6 _{±0.4}	96.1 _{±0.4}	3.8 _{±0.5}	0.41 _{±0.05}	0.074 _{±0.008}
B4 Majority vote	5.3 _{±0.4}	96.3 _{±0.4}	3.6 _{±0.4}	0.45 _{±0.05}	0.069 _{±0.008}
B5 Label-agreement	<u>5.1_{±0.4}</u>	<u>96.5_{±0.4}</u>	<u>3.4_{±0.4}</u>	<u>0.48_{±0.05}</u>	<u>0.063_{±0.007}</u>
B6 MultiPhishGuard	5.7 _{±0.5}	96.0 _{±0.4}	3.9 _{±0.5}	0.42 _{±0.05}	0.090 _{±0.010}
PHISHPROOF (GEA)	3.3_{±0.3}	98.4_{±0.3}	1.6_{±0.3}	0.71_{±0.04}	0.031_{±0.005}
Δ vs best baseline	−35%	+1.9	−53%	+48%	−51%

for every method. The vision-language model and perceptual logo tool run on a single commodity GPU; DOM, TLS, redirect, and calibration code are CPU-only. We log prompts, raw agent outputs, value-normalized nodes, tool results, thresholds, and per-page trust scores, and release the schemas, prompts, tool wrappers, calibration code, and figure-generation scripts. Because this draft reports call-count cost rather than wall-clock latency, the hardware is not used to support a numeric latency claim; the artifact release will include the exact hardware and software versions for the final run.

6.2 Selective detection

Risk–coverage and deployment coverage. We first ask whether scoring agreement on evidence improves the core selective trade-off. Tab. 4 reports the headline comparison and Fig. 3 the full curves. PHISHPROOF lowers AURC by 35% relative to the strongest baseline and roughly halves the false-positive rate at 80% coverage; most usefully for deployment, it raises the coverage attainable at 99% selective accuracy from 0.48 to 0.71, meaning an operator can act automatically on far more traffic at the same error budget. The mechanism is visible in Fig. 3: all methods meet at the shared full-coverage panel error – PHISHPROOF does not change the majority label, only the order in which pages are trusted – but its curve falls fastest as coverage tightens, because grounded evidence-agreement ranks the safe pages above the fragile ones more sharply than label concordance does. The label-aggregation baselines (B4, B5) edge out single-model confidence but remain well above PHISHPROOF, confirming that the gain comes from *where* agreement is measured, not merely from using several models.

6.3 Right-label, wrong-reason

Partitioning correct verdicts by grounded agreement. PHISHPROOF’s premise is that some correctly labeled pages are untrustworthy because the panel agrees on the label but not on grounded evidence. We test this by partitioning the correctly labeled pages with PHISHPROOF’s two factors (Tab. 5). About 11% fall in the right-label, wrong-reason region and a further 6% rest on ungrounded (shared-hallucination) evidence. These regions are not benign: under a small evidence-targeted perturbation their labels flip 41.6% and 47.9% of the time, against 6.3% for high-GEA pages – they are exactly the pages a minor change to the cited cue overturns. Yet the baselines act on them confidently, with mean single-model

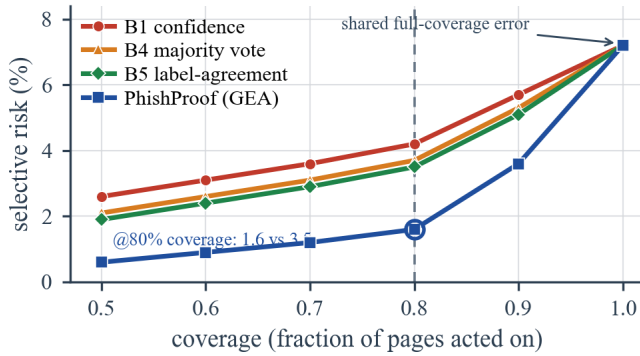


Figure 3: Risk-coverage on PHISHSEL (RQ1). Lower selective risk is better; the ring marks 80% coverage. PHISHPROOF changes the abstention order, not the majority label, and dominates across coverage.

Table 5: RQ2. Correctly labeled pages partitioned by evidence agreement and groundedness. “Flip” is label-flip rate under evidence-targeted perturbation. PHISHPROOF abstains on fragile RLWR and ungrounded regions.

Region (of correct labels)	Share	B1 conf.	Unanim.	Abstain	Flip↓
High-GEA ($A \uparrow$, $G \uparrow$)	72%	0.93	0.89	4%	6.3%
RLWR ($A \downarrow$ on evidence)	11%	0.91	0.64	78%	41.6%
Ungrounded ($G \downarrow$, shared halluc.)	6%	0.88	0.71	83%	47.9%
Low label-agreement (uncertain)	11%	0.74	0.18	81%	33.2%

confidence 0.91 and a unanimous panel label on 64% of the RLWR region; label-level reliability cannot see the problem because the labels do agree. PHISHPROOF abstains on 78–83% of these pages while acting on the stable high-GEA region, and Fig. 4(b) shows the same effect continuously: adversarial fragility falls monotonically from the lowest to the highest PHISHPROOF decile.

6.4 Component attribution

Per-component contribution. Tab. 6 removes one component at a time. Replacing evidence-level agreement with label agreement is the most damaging change: AURC rises to 5.0, essentially reproducing the B5 baseline of Tab. 4 and confirming that the contribution is the evidence channel, not the panel. Dropping groundedness admits shared hallucinations and costs the next most, and collapsing the panel to three same-family variants – removing diversity while keeping the count – erases much of the remaining advantage, which is why diversity, not panel size, is the design variable. Removing oracle calibration leaves ranking nearly intact but degrades the trust score’s ECE from 0.031 to 0.071, i.e. it mostly hurts the *reported* probability rather than the *order*. The groundedness tools are themselves accurate – the structural checks are exact by construction, and the perceptual logo tool attains 0.96/0.93 precision/recall against a gold logo-equivalence set – so G seldom discards a true node.

Table 6: RQ3. Component ablation. Underlined AURC values are at least 10% worse than full PHISHPROOF. Cells are mean $\pm$ 95% CI over five seeds.

Variant	AURC↓	FPR ₈₀ ↓	ECE↓
PHISHPROOF (full)	3.3$\pm$0.3	1.6$\pm$0.3	0.031$\pm$0.005
– calibration (R4)	3.6 $\pm$ 0.3	1.8 $\pm$ 0.3	0.071 $\pm$ 0.008
– diversity (same-family panel)	<u>4.4$\pm$0.4</u>	2.5 $\pm$ 0.4	0.044 $\pm$ 0.006
– groundedness G (R3)	<u>4.6$\pm$0.4</u>	2.7 $\pm$ 0.4	0.039 $\pm$ 0.006
– evidence-level A (R2)	<u>5.0$\pm$0.4</u>	3.3 $\pm$ 0.4	0.061 $\pm$ 0.007

Table 7: RQ4. Temporal robustness: in-period calibration/testing versus calibrating on months 1–3 and testing on months 4–6. Cells are mean $\pm$ 95% CI over five seeds.

Setting	PHISHPROOF (GEA)		B5 Label-agreement	
	AURC↓	ECE↓	AURC↓	ECE↓
In-period	3.3 $\pm$ 0.3	0.031$\pm$0.005	5.1 $\pm$ 0.4	0.063 $\pm$ 0.007
Temporal (drift)	3.7 $\pm$ 0.4	0.041$\pm$0.006	5.6 $\pm$ 0.5	0.079 $\pm$ 0.009

6.5 Calibration and drift

Reliability and temporal robustness. A selective detector is only as useful as its operating point, so we check whether the calibrated score is faithful and whether it survives drift. Fig. 4(a) plots reliability: PHISHPROOF’s calibrated trust tracks the diagonal, whereas single-model confidence sits below it – systematically over-confident – consistent with the ECE column of Tab. 4. To test drift we calibrate on the first three months and evaluate on the last three, a temporal protocol in the spirit of drift-aware malware evaluation [19] (Tab. 7). Both methods degrade as campaigns evolve, but PHISHPROOF retains its margin (AURC 3.3 $\rightarrow$ 3.7 versus 5.1 $\rightarrow$ 5.6 for B5), because grounded evidence-agreement depends on page-level cues that move more slowly than the surface features a label classifier keys on.

6.6 Cost and scope

Overhead and selective escalation. PHISHPROOF has no training phase and, in the measured configuration, runs $M=3$ agents per page. The extra work beyond those model calls is the verification of consensus nodes: DOM, TLS, and redirect lookups for structural units and one embedding comparison for a logo unit. We therefore report cost as an analytical call-count budget rather than as wall-clock latency, and read the resulting trade-off against quality in Fig. 5. Because grounded evidence-agreement reuses the single $M=3$ panel pass and adds only the cheap structural tool calls above, PHISHPROOF sits on the cost-quality Pareto frontier: it matches the call budget of the panel baselines (B4, B5) while reaching markedly higher deployable coverage, and it dominates the five-sample self-consistency baseline (B3) at lower cost. Selective operation also limits downstream escalation: at the 80% operating point, the abstaining fifth of traffic is sent to deeper analysis or a human, so downstream cost is governed by the same threshold that sets the risk-coverage point. A two-stage variant can first run a single agent and convene the full panel only for low-confidence pages; this is a deployment knob, not a separately measured result in this draft.

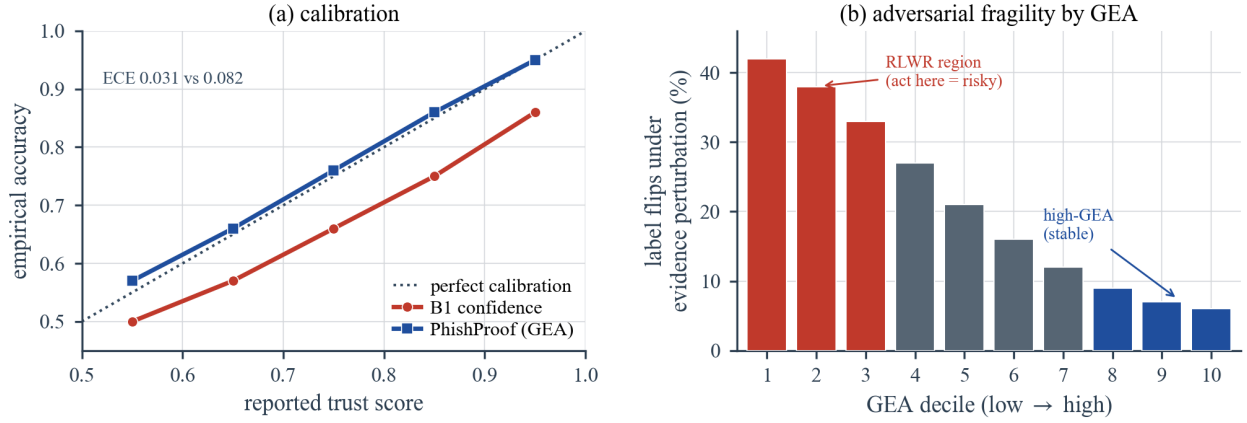


Figure 4: Diagnostics. (a) Calibrated trust tracks the diagonal while single-model confidence is over-confident. (b) Evidence-targeted label flips fall with GEA decile, showing that low-GEA/RLWR pages are fragile.

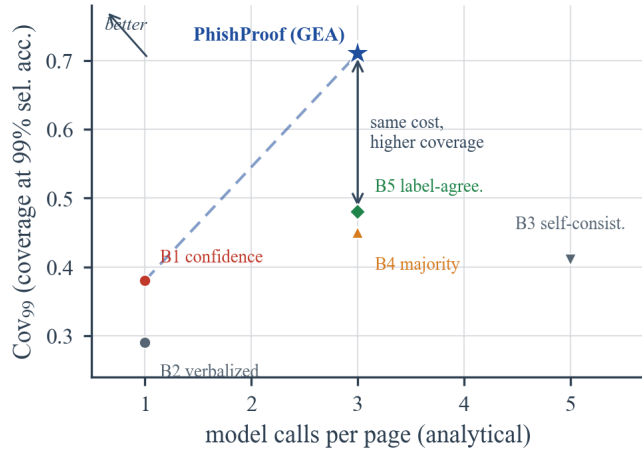


Figure 5: Cost-quality trade-off (RQ5): per-page model calls (analytical) versus coverage at 99% selective accuracy (Cov₉₉, ↑). PHISHPROOF is Pareto-optimal – at the panel’s call budget it reaches the highest deployable coverage and dominates the label-level and self-consistency baselines.

Measured scope. §5 discusses unseen brands and new evidence channels because the schema supports those extensions. We do not report their quality–cost trade-offs here, so they remain future-work settings rather than experimental claims.

7 Conclusion

Deployed phishing detectors are trusted at the label and explained, if at all, after the fact, even though the action a verdict triggers needs a reason an operator can check. We have argued that reliability should be read from the evidence rather than the label, and we instantiated this as PHISHPROOF: a multi-agent detector whose evidence agents investigate the page with tools and assemble a shared evidence graph, and whose Grounded Evidence-Agreement score trusts a verdict in proportion to how much independent agents agree on a subgraph that the page itself confirms, returns that subgraph

as the explanation, and abstains otherwise. Three findings stand out. The right-label, wrong-reason page – correctly labeled but resting on disagreed or hallucinated evidence – is a material part of the input space and the most adversarially fragile region, yet label confidence and majority voting act on it. Measuring agreement on evidence rather than on labels is what carries the improvement: PHISHPROOF lowers AURC by 35% relative to the strongest label-agreement baseline and raises coverage at 99% selective accuracy from 0.48 to 0.71. Because the score is grounded in page-level cues, it stays calibrated and keeps its margin as campaigns drift. Trust and explanation, in short, can be decoupled from raw accuracy and earned per verdict.

For future work, several extensions lie beyond the measured scope of this paper. A tool is sound for what it checks but a page can carry true-yet misleading evidence, so strengthening tools toward semantic plausibility would tighten the grounding guarantee. The evidence schema and the agent panel are fixed here; learning new evidence types, richer relations in the evidence graph, and ways to estimate true panel diversity would let the method adapt to new impersonation tactics and opaque third-party model pools. Because the calibrating oracle arrives late, the hardest regime remains the fully cloaked, novel page on which the detector can today only abstain; narrowing that gap with faster, page-internal grounding is the natural next step.

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